

DA3408 Assignment 1

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DA24B002

Question 1

The hidden-technical-debt categories for the given three situations are as follows:

1. CACE explains this the most. CACE (Change Anything Change Everything) claims altering one feature or hyperparameter (estimated delivery time rounding logic in this case) unexpectedly and silently shift the whole model's behaviour (changes favourite restaurants features).
2. Undeclared consumers explains this the most. Other teams (marketing dashboard team here) consume a model's output via a shared table, creating the invisible dependencies that block safe changes from the ML team's side.
3. Configuration Glue-Code debt explains this the most. Hyperparameters scattered across undocumented shell scripts with no single source of truth causes systems to evolve into tangled "pipeline jungles"

For part (c) where the entire training pipeline is made of unorganised shell scripts, we can use a software like mlflow to organise it. By specifying the pipeline declaratively, we can make sure all dependencies, environments, artifacts, parameters, execution orders, are documented in the code, making it easier to work together and keep track of previous code executions, and make the job much easier to orchestrate.

Question 2

The run with the parameters `hidden_layer_sizes = (256, 128)`, `learning_rate_init = 0.001` performed the best with the least training loss and the highest validation accuracy. The run with the parameters `hidden_layer_size = (128,)`, `learning_rate_init = 0.001` also achieved the same validation accuracy with a higher training loss.

There are mild signs of overfitting when the network size increases, as seen when training loss decreases manyfolds while validation accuracy stays roughly the same, as seen in `learning_rate_init = 0.001`.

I think `learning_rate_init` had the highest impact when it came down to model performance, as you can see the model performed really bad when `learning_rate_init = 0.0001`, and we got the optimal performance when `learning_rate_init = 0.001` as described previously

Run details						
Run ID:	3a22058174614d13d48cead0...	57414ad73ed94ba3be151406e...	9780c571cdc4ef6ad579e1a011...	8fce9e1178ee4830a282c8d115...	83014e5ee2842489b3be405d...	8d7279c9c374800a5434e4e41...
Run Name:	mfp_run_1	mfp_run_6	mfp_run_5	mfp_run_4	mfp_run_3	mfp_run_2
Start Time:	08/30/2026, 02:06:26 PM	08/30/2026, 02:09:16 PM	08/30/2026, 02:08:56 PM	08/30/2026, 02:08:02 PM	08/30/2026, 02:07:40 PM	08/30/2026, 02:06:47 PM
End Time:	08/30/2026, 02:06:47 PM	08/30/2026, 02:10:06 PM	08/30/2026, 02:09:16 PM	08/30/2026, 02:08:56 PM	08/30/2026, 02:08:02 PM	08/30/2026, 02:07:40 PM
Duration:	21.3s	50.1s	21.4s	52.2s	22.3s	52.9s
Parameters						
Show diff only						
hidden_layer_sizes	(128,)	(256, 128)	(128,)	(256, 128)	(128,)	(256, 128)
learning_rate_init	0.0001	0.01	0.01	0.001	0.001	0.0001
max_iter	20	20	20	20	20	20
Metrics						
Show diff only						
train_loss	0.147	0.051	0.038	0.004	0.012	0.054
val_accuracy	0.954	0.976	0.973	0.976	0.976	0.972

Question 3

```
shower@x-8 sujal sujal 4050 Aug 30 15:54 ~$ git /
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ find data -type f | wc -l
1800
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ dvc add data/
100% Adding...
```

To track the changes with git, run:

```
git add .gitignore data.dvc
```

To enable auto staging, run:

```
dvc config core.autostage true
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ git add .gitignore data.dvc
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .gitignore
    new file:   data.dvc
```

```
-rw-rw-r-- 1 sujal sujal 22958744 Aug 30 15:58 new-labels.zip
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ find data -type f | wc -l
2800
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$
```

```
data/train/dogs/dog_989.jpg
data/train/dogs/dog_990.jpg
data/train/dogs/dog_991.jpg
data/train/dogs/dog_992.jpg
data/train/dogs/dog_993.jpg
data/train/dogs/dog_994.jpg
data/train/dogs/dog_995.jpg
data/train/dogs/dog_996.jpg
data/train/dogs/dog_997.jpg
data/train/dogs/dog_998.jpg
data/train/dogs/dog_999.jpg

Modified:
data/

Files summary: 1800 added
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ dvc checkout
Building workspace index
Comparing indexes
Applying changes
+ data/
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ find data -type f | wc -l
1800
(allops) sujal@ideapad-3-151TL6:~/alops_da24b002$ dvc st
```